

THE EFFECT OF IRON CORE STRUCTURE ON SUPERNOVAE¹

E. BARON

Department of Physics, State University of New York at Stony Brook

AND

J. COOPERSTEIN

Department of Physics, Brookhaven National Laboratory

Received 1989 July 28; accepted 1989 October 18

ABSTRACT

We study the effects of the initial models on the outcome of stellar collapse calculations. Constructing our own initial models which satisfy constraints taken from the results of presupernova evolution calculations, we discuss the basic criterion that any initial model must satisfy. Nonequilibrium transport of all flavors of neutrinos and antineutrinos are included as well as the effects of neutrino electron scattering. We make conservative choices for the equation of state at high density. Cold initial cores significantly reduce the amount of electron capture on infall, leading to strong shock waves. We also find that there appears to be an optimum iron core mass and that reducing the mass below this optimum leads to increased lepton loss on infall and subsequent deleterious effects on the shock wave. We obtain successful explosions with iron cores of $M \sim 1.12 M_{\odot}$ when they have the proper characteristics.

Subject headings: dense matter — equation of state — neutrinos — stars: interiors — stars: supernovae

1. INTRODUCTION

Massive stars ($M \gtrsim 10\text{--}12 M_{\odot}$) become catastrophically unstable when the fuel in their central regions is exhausted. The innermost $1\text{--}2 M_{\odot}$ of fusion ashes (iron peak elements burnt to nuclear statistical equilibrium [NSE]) is the *iron core*. It resembles a white dwarf star dominated by the pressure of relativistic electrons, albeit a hot one. An isolated white dwarf cannot support more than the appropriate Chandrasekhar mass for its composition, but the compact core of the massive star must support the overlying burning shells. Before it evolves completely to the low-temperature white dwarf configuration, it loses this ability, is overwhelmed by gravitation, and collapses. It is generally agreed that a Type II supernova explosion ensues, but there is no general agreement about the details of the process. In fact, at present there exist no self-consistent calculations of the explosive stage which adequately explain observed explosion features.

There are two well-developed theories of the central moments of the explosion. In the first, the prompt mechanism (Arnett 1983; Baron 1985; Baron, Cooperstein, and Kahana 1985*a, b*; Bowers and Wilson 1982; Bruenn 1985, 1989*a, b*; Burrows and Lattimer 1985; Colgate and Johnson 1960; Cooperstein, Bethe, and Brown 1984; Hillebrandt 1982; Mazurek, Cooperstein, and Kahana 1980; Myra *et al.* 1987; Myra and Bludman 1989; Takahara and Sato 1982, 1984, 1985; Van Riper 1978, 1979, 1980, 1982; Wilson 1980), the star implodes to supranuclear densities and then resists further compression. The inner core rebounds and launches a shock wave which then propagates through the star and eventually erupts through the photosphere producing the supernova display. In the second theory, the delayed mechanism (Bethe and Wilson 1985; Mayle 1985; Mayle and Wilson 1988; Wilson 1985), there is insufficient energy in the initial shock wave to propagate all the way to the edge of the star. Instead, it

falters and becomes an accretion shock. However, about 1 second after the central bounce, it is revived by deposition of a small percentage of the energy leaving in neutrinos, and it resumes its outward motion. The result is a weaker explosion than in the prompt mechanism. In this paper, we will focus on the prompt mechanism.

Whether or not the prompt shock can succeed is a sensitive function of both the initial configuration at the onset of instability and of the physics used to follow the hydrodynamical evolution onward. While the elementary aspects of some ingredients, such as the neutrino interactions, are well enough known, the actual results of different investigations have to some degree been dependent on the details of the methods of calculation employed. However, other ingredients, such as the precise nature of the equation of state of material at very high density, are not as well known and contain large uncertainties which impact upon the outcome of hydrodynamical simulations.

In this paper, we will make conservative assumptions about these physical uncertainties during the explosive stage (i.e., we will adopt parameter values and physical inputs which are unfavorable to the prompt mechanism, such as a “stiff” equation of state and inclusion of neutrino-electron scattering), and we will explore what is required of an initial model to obtain success. One exception to our conservative assumptions is the lack of any electron capture reactions on nuclei; there is no doubt that proper inclusion of both allowed and forbidden electron capture reactions would be harmful to producing successful explosions.

It has been repeatedly observed that the mass of the iron core is one of the most important inputs into the hydrodynamical simulations. Because the principal dissipative loss the shock wave suffers is the energy it must supply to dissociate infalling nuclei of the iron core into free nucleons, reducing the iron core mass helps enormously (Baron 1985; Burrows and Lattimer 1983; Cooperstein, Bethe, and Brown 1984), however, we shall see that the iron core mass, while a critical parameter,

¹ This work has been supported in part by the US Department of Energy under contract no. DE-AC02-76CH00016 and grant DE-FG02-88ER40388.

is not the only one; we will construct iron cores of the same nominal mass which have quite different qualities (entropy, composition, and density profiles) which make a large difference to the viability of the explosion.

II. S AND Y_e PROFILE OF THE IRON CORE

The most satisfactory initial models for hydrodynamical collapse are the final models evolved by late stellar evolution codes. However, there are considerable uncertainties involved and indeed, there has been much variation in these results over the last decade. Thus, it has often been advantageous to construct “fake models” to explore the possible outcomes. This can be done either by fiddling with the results of the presupernova evolution to explore variations (Bruenn 1989b), or by direct construction (Baron 1985; Cooperstein 1982; Cooperstein, Bethe, and Brown 1984). We will follow the second approach because it is more consistent. The hope is that the complicated results of the presupernova evolution can be mapped onto only a few quantities whose profiles can be varied.

There are three significant variables which must be specified as a function of mass. The first two are the entropy and composition (Y_e) profiles. The third is the acceleration (i.e., the pressure deficit), because the models are already collapsing quickly by the time they are delivered to the simulations.

Silicon burning proceeds through several stages which vary among calculations and helium core mass (see Nomoto and Hashimoto 1988; Woosley and Weaver 1988). There is always a period of convective central silicon burning, and then it is followed by zero, one, or two shell flashes which increase the mass of the NSE core. After (and during) each burning stage, the material cools by neutrino emission through pair production and also via β -reaction generated neutrinos. These latter reactions change the Y_e profile as well. Both the cooling and the neutronization proceed most rapidly at the center, and so the result is that material which begins to burn convectively with uniform S and Y_e develops positive gradients in these quantities, because the center is depressed more quickly than the outer regions. Taking M_1 to be the mass of the central formerly convective region, we incorporate these features by writing for $M \leq M_1$,

$$S(M) = S_c + (S_1 - S_c) \left(\frac{M}{M_1} \right)^{n_s}, \quad (1)$$

$$Y_e(M) = Y_{ec} + (Y_{e1} - Y_{ec}) \left(\frac{M}{M_1} \right)^{n_y}, \quad (2)$$

where S_c and Y_{ec} are the central values of the entropy and electron fraction and S_1 and Y_{e1} are the values at M_1 . The powers n_s and n_y control the curvature of the gradients. The mass-weighted average values of the compositional parameters of the inner, previously convective core are thus

$$\langle S \rangle_{M \leq M_1} = \frac{n_s S_c + S_1}{n_s + 1}, \quad (3)$$

$$\langle Y_e \rangle_{M \leq M_1} = \frac{n_y Y_{ec} + Y_{e1}}{n_y + 1}. \quad (4)$$

It turns out that it is really only these averages that control the initial structure of the profile as far as pressure and density gradients go, and so the simplest case ($n_s = n_y = 1$) will be adopted.

Figure 1 represents a schematic view of the S and Y_e profiles of an iron core which is undergoing shell-burning of silicon after the central convective core has been exhausted of fuel. At mass point M_1 , there will be a rise in the entropy as we reach either the shell-flashed silicon burnt material, or in the case where there is no shell flash, the silicon shell. We will assume that this rise takes place over a finite mass range (which we may take to be quite small to represent a discontinuity) and that in this region the entropy rises linearly to mass point M_2 . We will assume that in this next shell, the entropy is constant at $S_2 = S_3$ out to mass point M_3 (at which point it undergoes a gradual but sharp rise to S_4 at M_4) and that past that point, the entropy remains constant. For a comparison of the schematic with realistic calculations, Figure 2 plots the S , Y_e , T , and ρ profiles of the $13 M_\odot$ model of Nomoto and Hashimoto (1988), the $15 M_\odot$ model of Woosley and Weaver (1988), the $18 M_\odot$ model of Weaver and Woosley (1988), and the “benchmark” model used in calculations 102, 103, and 106. Thus, if we assume all rises are linear with mass for simplicity, our full S profile becomes as follows:

$$\begin{aligned} S(M \leq M_1) &= S_c + (S_1 - S_c) \left(\frac{M}{M_1} \right), \\ S(M_1 \leq M \leq M_2) &= S_1 + (S_2 - S_1) \left(\frac{M - M_1}{M_2 - M_1} \right), \\ S(M_2 \leq M \leq M_3) &= S_2 = S_3, \\ S(M_3 \leq M \leq M_4) &= S_3 + (S_4 - S_3) \left(\frac{M - M_3}{M_4 - M_3} \right), \\ S(M \geq M_4) &= S_4. \end{aligned} \quad (5)$$

Observing the results of full supernova evolution calculations (see Fig. 2), we see that the Y_e profile is somewhat simpler. The gradual rise in Y_e continues out to edge of the NSE core at which point it rises sharply. This is because the Y_e profile is mostly determined by the time since the material was burned to NSE, and the first shell burning begins almost immediately after core exhaustion, so there is no discontinuity. Thus, for the Y_e profile we obtain:

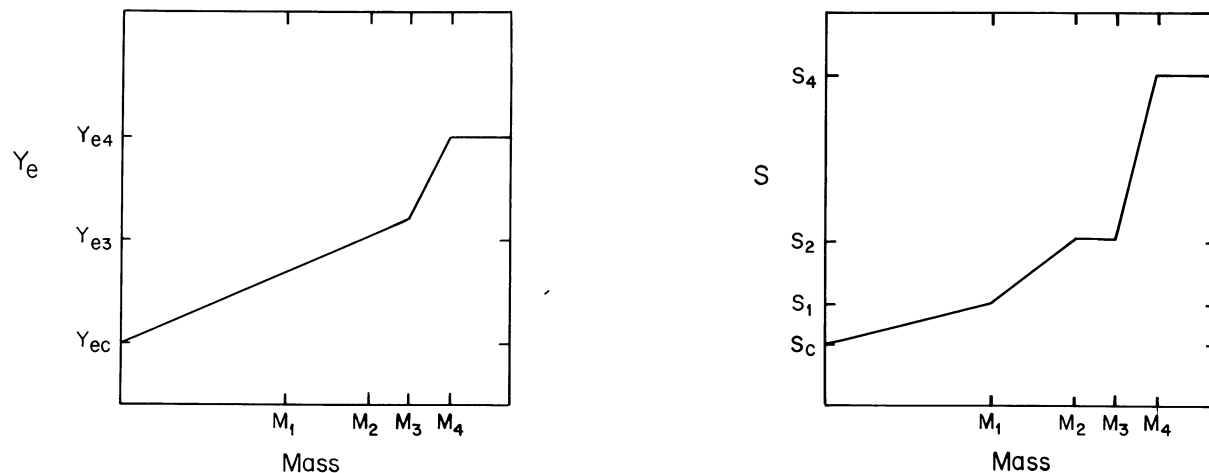
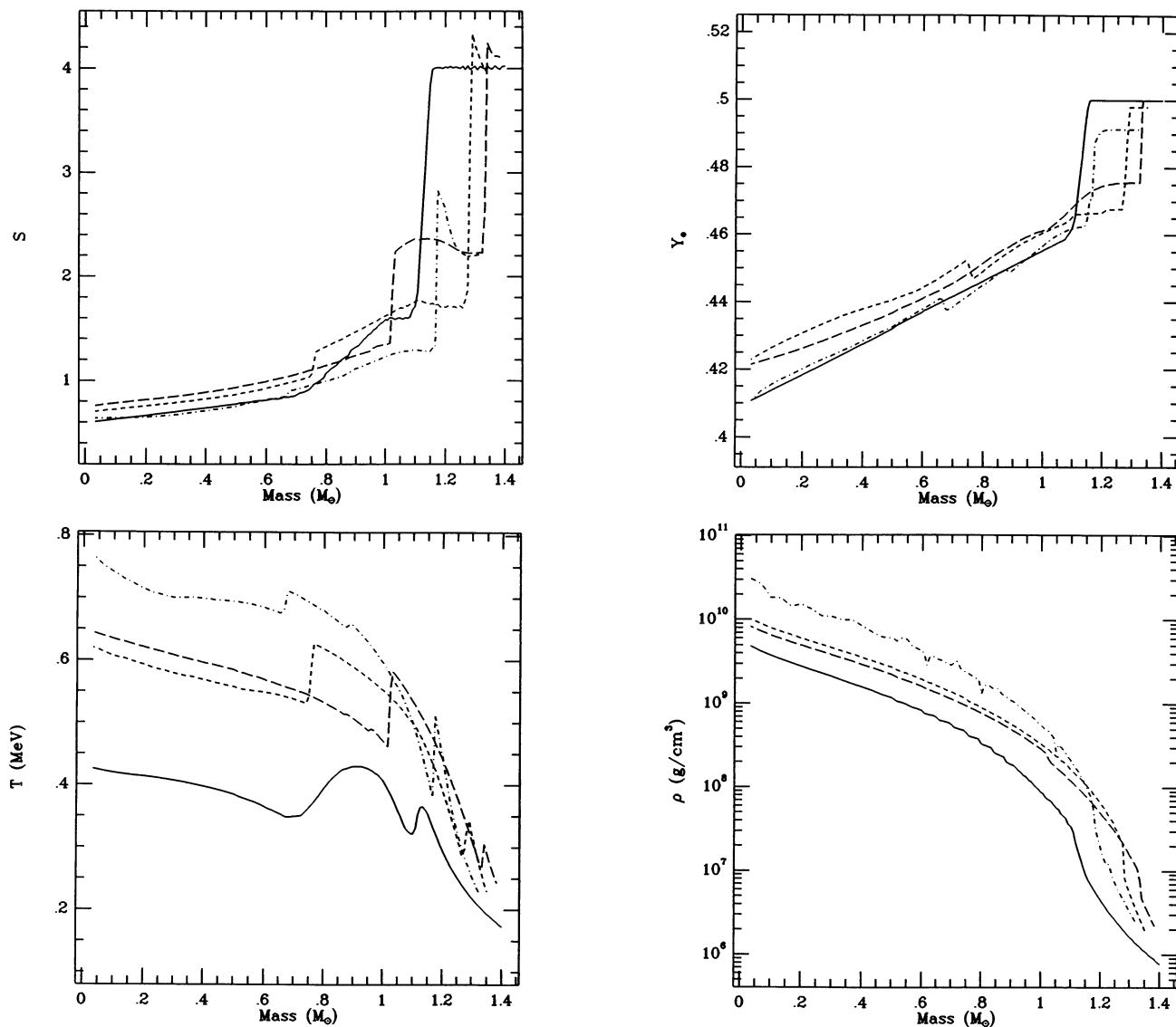
$$\begin{aligned} Y_e(M \leq M_3) &= Y_{ec} + (Y_{e3} - Y_{ec}) \left(\frac{M}{M_3} \right), \\ Y_e(M_3 \leq M \leq M_4) &= Y_{e3} + (Y_{e4} - Y_{e3}) \left(\frac{M - M_3}{M_4 - M_3} \right), \\ Y_e(M \geq M_4) &= Y_{e4}. \end{aligned} \quad (6)$$

For the mass-weighted average values out to the edge of the first silicon-flashed zone, we obtain:

$$\begin{aligned} \langle S \rangle_{M \leq M_3} &= \left(\frac{S_c + S_1}{2} \right) \left(\frac{M_1}{M_3} \right) + \left(\frac{S_1 + S_2}{2} \right) \\ &\quad \times \left(\frac{M_2 - M_1}{M_3} \right) + S_2 \left(\frac{M_3 - M_2}{M_3} \right), \end{aligned} \quad (7)$$

$$\langle Y_e \rangle_{M \leq M_3} = \left(\frac{Y_{ec} + Y_{e3}}{2} \right). \quad (8)$$

It is not hard to construct other models with these same averages with somewhat different details, such as one in which $M_1 = M_2$ (no shell flash), which would require an increase in

FIG. 1.—The schematic profiles of Y_e and S are displayedFIG. 2.—The profiles of S , Y_e , T , and ρ of our “benchmark” model (solid line) (used in calculations 102, 103, and 106) are compared to the pre-supernova calculations of the Woosley and Weaver (1988) 15 $M_\odot$ model (short dashed line), the 18 $M_\odot$ model of Weaver and Woosley (1988) (long dashed line), and the Nomoto and Hashimoto (1988) 13 $M_\odot$ model (dot-dashed line).

S_c and/or S_2 . We will give examples when we actually build our models and consider the dynamical consequences of them.

We have cast our discussion in terms of the entropy S , and not the temperature T , because once nuclear burning is over, S does not change during adiabatic compression or expansion, although it does drop due to neutrino cooling. However, nuclear burning depends very sensitively upon T and only weakly upon the density. In particular, there is an ignition temperature,

$$T_{\text{sil}} \sim 0.35 \text{ MeV} \sim 4.1 \times 10^9 \text{ K}, \quad (9)$$

for shell-burning silicon into NSE (Nomoto and Hashimoto 1988; Woosley and Weaver 1988), and it is reasonable to demand that for $M \geq M_4$, we have $T \geq T_{\text{sil}}$. This will narrow down the range of possible profiles. (The silicon ignition temperature at the center for the central convective region is actually somewhat less than $T = 0.35 \text{ MeV}$ because of its higher density, but the relevant ignition point for us is the higher value because we are concerned with shell burning in lower density shells.) Note, however, that T can drop below T_{sil} once material has burned due to neutrino cooling. At some point, the material will “freeze out” from nuclear statistical equilibrium.

In Table 1, we give the compositional parameters used for the fake models for which we will perform hydrodynamic simulations. Table 2 gives further parameters used to define these models, which will be elaborated upon shortly. Our “benchmark” model, used in calculations 102, 103, and 106, has $M_{\text{ch}} \sim 1.09 M_{\odot}$ and $M_{\text{iron core}} = 1.125 M_{\odot}$ (defined in the next section), and we will consider its properties in some more detail for illustration as representative of all the models. The selection of these models will be motivated in § V.

III. CHANDRASEKHAR AND IRON CORE MASSES

It is often stated (imprecisely) that the iron core becomes unstable when its mass rises above its Chandrasekhar mass. We will see that in fact the iron core mass can be less than the

TABLE 1
INITIAL MODEL COMPOSITIONS

PARAMETER	MODEL					
	102, 103, 106	104	105	107	108	109, 110
M_1	0.72	0.72	0.72	0.72	0.85	0.72
M_2	1.00	1.00	0.90	1.00	1.05	1.00
M_3	1.10	1.10	0.975	1.05	1.05	1.10
M_4	1.15	1.15	1.00	1.10	1.10	1.15
$M_{\text{iron core}}$	1.125	1.125	0.988	1.075	1.075	1.125
$M_{\text{ch}}^{(a)}$	1.089	1.115	1.065	1.090	1.090	1.102
$M_{\text{ch}}^{(b)}$	1.092	1.103	1.069	1.089	1.081	1.088
$M_{\text{ch}}^{(c)}$	1.087	1.088	1.036	1.087	1.087	1.057
S_c	0.60	0.50	0.60	0.60	0.60	0.50
S_1	0.86	0.70	0.86	0.86	0.91	0.63
S_2	1.60	1.60	1.30	1.60	1.60	1.60
S_4	4.00	4.00	4.00	4.00	4.00	4.00
$\langle S \rangle_{M \leq M_3}$	0.94	0.83	0.84	0.91	0.85	0.80
Y_{ec}	0.41	0.42	0.40	0.41	0.41	0.415
Y_{e3}	0.46	0.46	0.46	0.46	0.46	0.46
Y_{e4}	0.50	0.50	0.50	0.50	0.50	0.50
$\langle Y_e \rangle_{M \leq M_3}$	0.435	0.440	0.430	0.435	0.435	0.438

NOTE.—The meaning of the symbols is discussed in the text. Masses are given in units of $M_{\odot}$, and entropy is in k_B^{-1} per nucleon.

TABLE 2
MODEL PARAMETERS

Model	K_0	γ	W_s	NES	$\rho_{10}^{(c)}$	g_{eff}
102.....	180	3.0	31.5	off	0.538	0.975
103.....	180	3.0	31.5	on	0.538	0.975
104.....	180	3.0	31.5	on	0.500	0.975
105.....	180	3.0	31.5	on	0.150	0.950
106.....	180	2.5	31.5	on	0.538	0.975
107.....	180	3.0	31.5	on	0.250	0.975
108.....	180	3.0	31.5	on	0.336	0.975
109.....	180	3.0	31.5	on	2.000	0.975
110.....	180	2.5	31.5	on	2.000	0.975

NOTES.—The parameters used in each calculation. K_0 is the incompressibility of symmetric matter at saturation in MeV, and γ is the high-density adiabatic index; see eq. (17). W_s , the symmetry energy, is measured in MeV, NES refers to whether the effects of neutrino electron scattering are included, and $\rho_{10}^{(c)}$ is the initial central density in units of $10^{10} \text{ g cm}^{-3}$. g_{eff} is discussed in the text.

Chandrasekhar mass when it goes unstable, but that this apparent contradiction is probably more a reflection of the difficulty of appropriately defining either of these quantities than it is of any real inconsistency. The point is that the iron core has a finite value of the pressure at the edge, whereas the Chandrasekhar core has zero pressure. Since the iron core is being pushed at the edge, the mass at which it becomes unstable can be less than the Chandrasekhar mass. Ignoring Coulomb lattice corrections, the zero-temperature mass is given by

$$M_{\text{ch}}^{(a)} = 1.44 \left(\frac{Y_e}{0.5} \right)^2. \quad (10)$$

Exactly when the core becomes unstable is critical for the ensuing hydrodynamical events, and thus a closer look is warranted. A first refinement would be to note that the Chandrasekhar mass itself decreases in late stellar evolution, and so the crisis is forced from the confluence of the increase in its actual mass and decrease in its Chandrasekhar mass. However, each of these masses is difficult to define because the “white dwarf” we are analyzing is embedded in a giant star, and so a further refinement of the instability condition requires a more precise elucidation of these quantities.

The zero-temperature Chandrasekhar mass is modified by several corrections, the most important of which are the Coulomb lattice effects, the thermal pressure in electrons, and the thermal pressure in ions (including free nucleons and alpha particles). There are other corrections such as those due to nuclear excited states, radiation, general relativity, electron rest mass, and rotation, but for our conditions these are shown to be small in the Appendix. We evaluate the major corrections in the Appendix, and we find

$$M_{\text{ch}}^{(b)} = 1.44 \left(\frac{Y_e}{0.5} \right)^2 \left[1 - 0.057 + \left(\frac{S_e}{\pi Y_e} \right)^2 + 1.21 \left\langle \frac{1}{A} \right\rangle S_e \right] M_{\odot} \quad (11)$$

$$= 1.44 \left(\frac{Y_e}{0.5} \right)^2 \left(1 - 0.057 + 0.0013 T^2 \rho_{10}^{-2/3} + 0.680 \left\langle \frac{1}{A} \right\rangle T \rho_{10}^{-1/3} \right) M_{\odot}, \quad (12)$$

where ρ_{10} is the density in units of $10^{10} \text{ g cm}^{-3}$, and the entropy of electrons per nucleon is given by

$$S_e = Y_e \left(\frac{\pi^2 T}{\mu_e} \right) = 0.56 \left(\frac{Y_e}{0.50} \right)^{2/3} T \rho_{10}^{-1/3}, \quad (13)$$

with μ_e the electron chemical potential, and $\langle 1/A \rangle$ the mass fraction weighted average reciprocal atomic mass number, i.e., the reciprocal of the mean molecular weight. Strictly speaking, this formula only makes sense if the core is uniform so that the correction terms are independent of mass. While this is true for the first correction, the 5.7% reduction due to the negative Coulomb pressure, it is not true for the positive second and third corrections due to the thermal pressure of electrons and ions, respectively. However, sense can still be made by using properly averaged values over the core.

It is more convenient to use equation (11) than equation (12) because it requires only S_e and not ρ and T . However, our compositional profile specifies only the total entropy S , and how it is subdivided into its various components depends on all its thermodynamic variables. In general, at fixed S the ratio S_e/S increases with decreasing density for two reasons. First, as the temperature rises with density, more entropy is cast into nuclear excitation. Second, photodisintegration reactions begin to break up nuclei and throw more entropy into baryonic translational motion. One finds at the centers of our models $S_e/S \sim 0.3$. In the outer regions, we find $S_e/S \sim 0.6$, and at the mass point where $S = \langle S \rangle$, we find $S_e/S \sim 0.5$. Equation (11) requires $\langle S_e \rangle$, and we take $\langle S_e \rangle = \langle S \rangle/3$. This accounts for the fact that the Chandrasekhar mass should depend mostly on the central regions of the star where the pressure is highest, because stability considerations generally involve pressure-weighted integrals. The agreement between $M_{\text{ch}}^{(b)}$ and $M_{\text{ch}}^{(a)}$ shown in Table 1 is quite remarkable and will be commented upon after discussion of the third definition $M_{\text{ch}}^{(c)}$.

It is worthwhile to recall how the Chandrasekhar mass is defined. For a purely relativistic ($\gamma = 4/3$) gas, Newtonian polytrope analysis finds a unique mass, dependent upon only the pressure constant κ in the equation of state $P = \kappa \rho^\gamma$. The energy of such a configuration is identically zero (aside from an unimportant constant), and thus a fully relativistic star can be expanded or contracted without any cost; it is metastable. Since the electron equation of state approaches the exactly relativistic form only at infinite density, the Chandrasekhar limit is the limiting mass of an electron pressure-supported star in this high-density limit. For stars obeying equations of state not fully relativistic or uniform, such as a neutron star, it is still possible to define a generalized Chandrasekhar mass, and this can be done using the general relativistic hydrostatic structure equations instead of the Newtonian ones. For smooth equations of state, the maximum stable mass can be found by varying the central density until the gravitational mass (or, equivalently, the binding energy) is maximized. Stars with higher central densities are unstable. In all cases, a compositional profile must be used, and the Chandrasekhar mass spoken of is the one for that profile.

In finding the Chandrasekhar mass, one examines a series of models for which one integrates the hydrostatic structure equations until the pressure vanishes, a condition which defines the edge of the star. However, the iron core has a sharp entropy rise in its outer layers which permits it to support the overlying giant star. Mathematically, this means there is an external boundary condition which differs from the $P = 0$ of

the isolated white dwarf, which describes the white dwarf-like iron core. In our fake models, we have supplied as an outer boundary condition that $T = T_{\text{sil}}$ at M_4 . Since we are also specifying S and Y_e at this mass point, we are also specifying its density and pressure.

A third definition for the Chandrasekhar mass, $M_{\text{ch}}^{(c)}$, based on practical considerations, can be obtained numerically and is probably the most relevant quantity. To evaluate it, we extrapolate the S and Y_e profiles interior to M_1 (the pre-NSE convective core) past M_1 , peeling off the burning shells. The core is then unable to support the overlying giant star, and the pressure vanishes at a finite mass which varies slowly with central density. The central density which maximizes the total mass varies somewhat with each specific configuration, as it should, but in all cases it is quite close to $\rho_{10}^{(c)} \sim 7$. We label the mass thereby ascertained as $M_{\text{ch}}^{(c)}$ and list it in Table 1. We find $M_{\text{ch}}^{(c)} \sim M_{\text{ch}}^{(a)}$, the zero-temperature uncorrected mass for the average Y_e , meaning the positive thermal and negative Coulomb lattice corrections cancel almost perfectly. For most of the models, the cancellation is good to a few tenths of a percent. For model 105, the numerical estimate is 3% smaller, and for model (109, 110), it is 4% smaller than the simplest analytical one. It is perhaps not surprising that the different methods give similar results, since the corrections are roughly of the same order. It is probably most useful to compare different models using one definition.

The mass point where we terminate the “iron core” is also somewhat ambiguous. At $M = M_3$, the entropy begins to rise, and at $M = M_4$, it has reached the flat entropy level of the silicon burning shell. In realistic models such boundary regions are also seen, as can be observed in Figure 2. Thus, for convenience and need of a consistent rule, we will take the following definition:

$$M_{\text{iron core}} = \frac{1}{2}(M_3 + M_4). \quad (14)$$

IV. DESTABILIZATION AND COLLAPSE

While we can specify the S and Y_e profiles with some freedom, there is no guarantee that any given profile can support the overlying regions of the star. If we assume that the star is in hydrostatic (or at least quasistatic) equilibrium and select a central density and the profiles, we can integrate outward using the hydrostatic structure equations. If the density drops precipitously before we reach M_3 , it will be impossible for the specified object to support the giant star in which it is embedded. It turns out that while formally this must also depend on the central density, practically this is not true.

In addition to the thermal and compositional profile, the state of dynamical collapse must be specified to construct realistic “fake” models. This can be done in a number of ways. The usual method is to construct a stable hydrostatic model first, by using a somewhat “stronger” profile than required; i.e., by using a higher S , T , or Y_e , and then destabilizing this model by reducing the augmented quantity, but retaining the radius of the mass elements in the augmented stable star. The pressure will no longer be sufficient to support the star, and then collapse ensues. One can put in an initial velocity field as well, but numerical experiments show that this field is only a transient which is rapidly washed out as the material settles into homologous collapse.

In this approach, the most commonly adopted method is to reduce Y_e while keeping T fixed (not S), because the pressure is dominated by relativistic electrons. For example, a 5% drop in

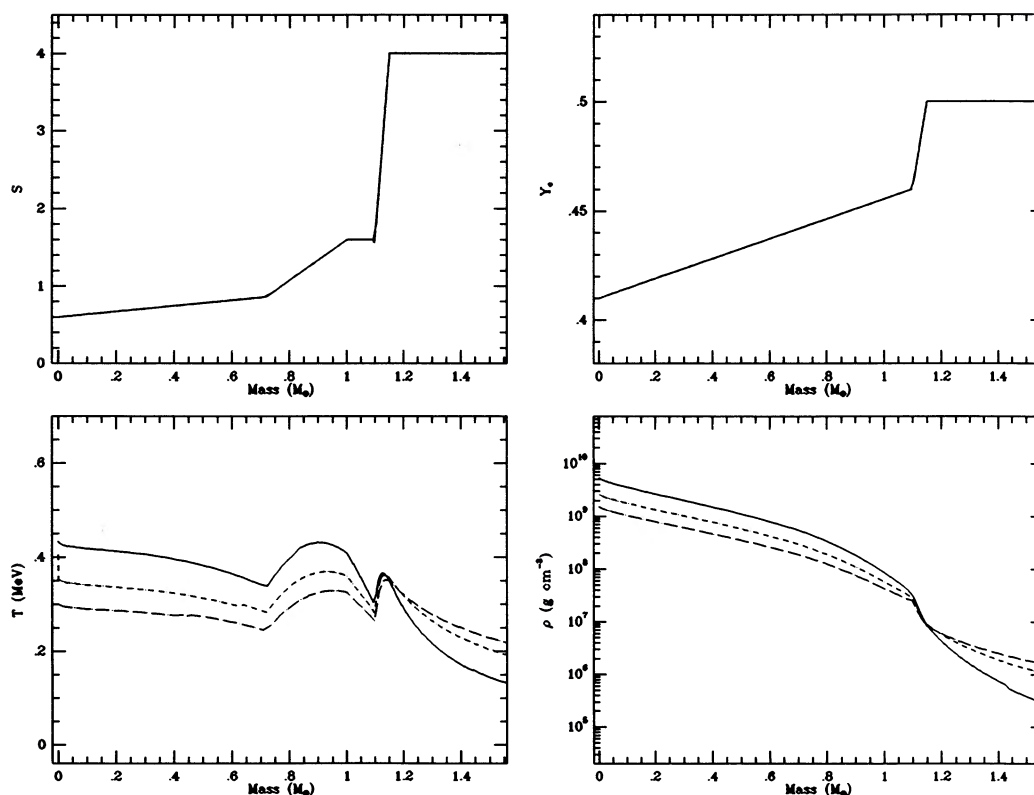


FIG. 3.—The effect of varying g_{eff} while keeping the temperature at M_4 fixed. The solid line has $g_{\text{eff}} = 0.975$, while the short dashed line is for $g_{\text{eff}} = 0.950$ and the long dashed line is for $g_{\text{eff}} = 0.925$.

Y_e leads to a 6.7% drop in the pressure, since $P \propto Y_e^{4/3}$, and a 10% drop in the relevant Chandrasekhar mass which obeys $M_{\text{ch}} \propto Y_e^2$. This is the approach followed in our earlier investigations with artificial models (Baron 1985; Baron, Cooperstein, and Kahana 1985b; Cooperstein 1982; Cooperstein, Bethe, and Brown 1984), but it suffers some deficiencies. Because the material is not completely degenerate ($T, S > 0$), changing Y_e alters the entropy in a complicated way. Thus, even if Y_e is reduced uniformly in the core, there will be a nonuniform pressure reduction. While there is no deep reason why the destabilization should be uniform, the fact is that in the initial models delivered to the hydrodynamical simulations by the presupernova stellar evolution codes, the destabilization as measured by the relative pressure deficit out to the edge of the iron core is fairly uniform. This is because the destabilization is a product of both electron captures and photodisintegration, with electron capture being most important at the center and photodisintegration being most important at the edge of the iron core, in such a way that the net effect is quite uniform.

As an improvement, we adopt the following procedure. We pretend that Newton's constant G can be varied; i.e., we take

$$G = g_{\text{eff}} G_0, \quad (15)$$

where G_0 is the actual value and $g_{\text{eff}} < 1$. Selecting a central density, we then build a model utilizing a general relativistic hydrostatic code with the same equation of state we will use in the hydrodynamics. At each mass point, there will be a resulting acceleration given by

$$\frac{dv}{dt} = -(1 - g_{\text{eff}}) \frac{G_0 M}{r^2}, \quad (16)$$

aside from quantitatively unimportant general relativistic corrections. The S and Y_e profiles are unaltered in this procedure. The central density is then iterated until the temperature at the base of the silicon burning shell, $T(M_4)$, converges to T_{sil} .

Which value of g_{eff} should one choose, and what effect does this choice have? Figure 3 shows the effects on our benchmark model, calculations 102, 103, and 106, of varying g_{eff} . For the values $g_{\text{eff}} = 0.975, 0.950$, and 0.925 , we vary the central density until $T(M_4) = T_{\text{sil}}$, and we thereby generate the runs of temperature and density with respect to mass. As g_{eff} is reduced, the value of the central density required is lowered; for $g_{\text{eff}} = (0.975, 0.950, 0.925)$, we find $\rho_{10}^{(c)} = (0.538, 0.250, 0.153)$. Likewise, the central temperature drops because we have fixed S_c and Y_{ec} . However, as the core evolves to higher density, one expects that g_{eff} should decrease as destabilization becomes more intense; indeed, in realistic presupernova calculations, infall velocities have reached 10^8 cm s^{-1} when $\rho_{10}^{(c)} \sim 1$. The central temperature drops well below T_{sil} (which in itself is greater than central ignition temperatures) to $T \sim 0.3 \text{ MeV}$. This is of course possible, but it has not been reflected in presupernova calculations to date (Nomoto and Hashimoto 1988; Woosley and Weaver 1988; Weaver and Woosley 1988). We conclude that for a large reduction in g_{eff} , the central density becomes too low to be consistent with the results of presupernova evolutionary calculations at the collapse point, and/or the central temperature has cooled far below the central silicon ignition temperature and the temperature gradient may be too flat.

Figure 3 shows that the situation in the outer regions of the iron core is somewhat different. Because T, S , and Y_e are fixed at M_4 , ρ is also. Since the density increases with g_{eff} inside of M_4 but has fixed value at this point, its gradient becomes more

negative at M_4 , and thus the outer regions will have lower density as g_{eff} is increased. This makes for more favorable shock propagation.

We argue that one should take a high value of g_{eff} (only a moderate destabilization), $g_{\text{eff}} = 0.975$. This produces as small an iron core as is possible for a given Chandrasekhar mass and gives the most physically reasonable temperature profile.

We can examine the effect of our outer boundary condition by relaxing it, and the results are displayed in Figure 4. Here we take $g_{\text{eff}} = 0.975$ and vary the central density and then display the runs of T and ρ as before. We see that except for the lowest densities (which we believe give inappropriate temperatures), the gradients in temperature and density are hardly affected at the outer edges of the iron core. We have also tested the sensitivity to the central density (and hence the outer boundary condition) numerically, by collapsing hydrodynamically some of the models shown in Figure 4 and comparing the results. At bounce graphs of the lepton fraction, entropy, velocity, etc., inside the developing shock front show absolutely no difference. The material simply “falls down” the adiabat until it reaches the same state as it would have from any initial density. There may be some effects on shock propagation in the outermost layers, and while it would be useful to investigate these effects, we already have many parameters to vary and they are at the same level as the uncertainties in our construction of the outer structure itself, so we will not pursue them further.

V. ANALYSIS OF MODELS AND CALCULATION PARAMETERS

In the previous sections, we have outlined our method of construction of initial models for collapse and oriented them

with respect to the Chandrasekhar mass. We have not yet discussed in detail the specific parameter choices used to construct them, which are given in Table 2. Some of the models will be used for more than one hydrodynamical calculation, using different physical assumptions regarding neutrino interactions or the equation of state. We discuss first the parameters which affect the initial model, at least formally.

A given profile of S and Y_e can produce somewhat different models based upon the exact equation of state employed. The equation of state used in all our calculations has been discussed elsewhere (Baron 1985; Baron, Cooperstein, and Kahana 1985*a, b*, hereafter BCK; Cooperstein 1985), recently reviewed in detail (Cooperstein and Baron 1989), and employed widely in hydrodynamical simulations (Baron 1985; Baron *et al.* 1987; BCK; Bruenn 1989*a, b*; Cooperstein and Baron 1989; Myra and Bludman 1989). For below nuclear saturation densities, the equation of state uses a nuclear statistical equilibrium description of a representative compressible liquid drop model nucleus in coexistence with a drip vapor of free nucleons and alpha particles. Electrons, photons, and neutrinos are included. The few adjustable parameters for the liquid drop description are: K_0 , the nuclear incompressibility; W_s , the bulk symmetry coefficient; ρ_s , the nuclear saturation density; and E_{nm} , the bulk saturation energy of nuclear matter. In addition, some information is built in about how K_0 and ρ_s depend on neutron excess, and a specific functional form is used for the nuclear partition function.

All the models discussed in this paper use $K_0 = 180$ MeV. This value is for symmetric matter; for twice as many neutrons as protons ($Z/A = \frac{1}{3}$), a ratio appropriate to the later stages of collapse, the BCK equation of state gives $K_0 = 140$ MeV. The

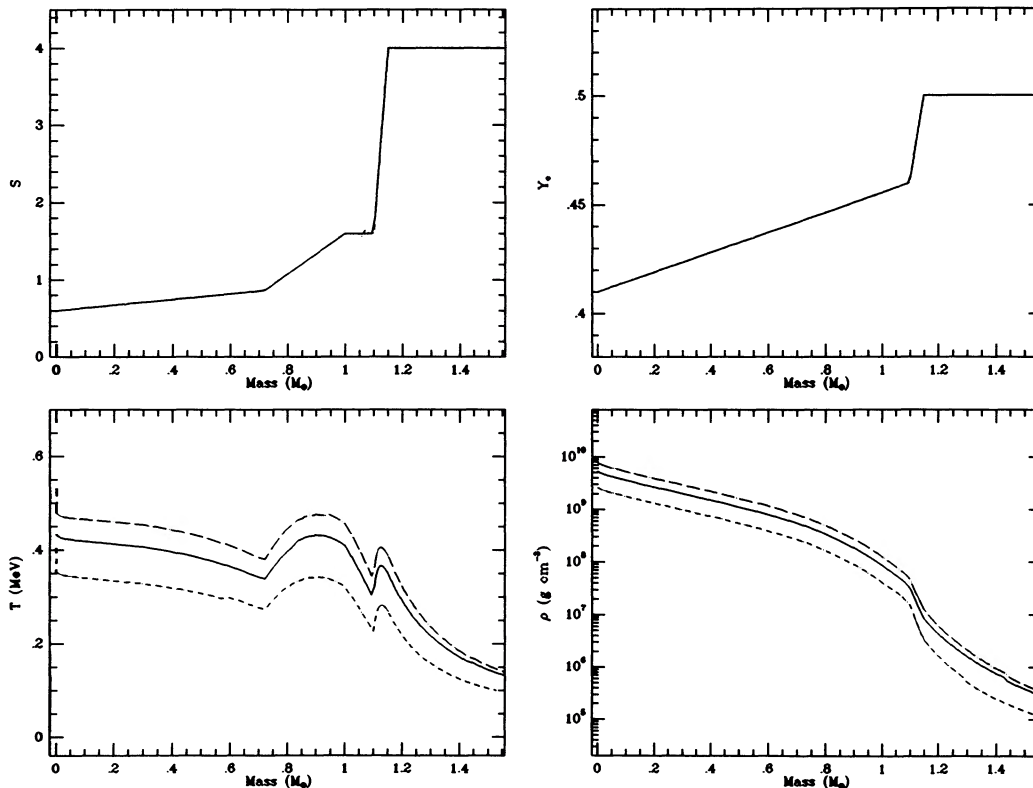


FIG. 4.—The effect of varying the central density. The solid line has $\rho_c = 5.83 \times 10^9 \text{ g cm}^{-3}$, our “benchmark model,” the short dashed line has $\rho_c = 2.0 \times 10^9 \text{ g cm}^{-3}$, and the long dashed line is for $\rho_c = 7.0 \times 10^9 \text{ g cm}^{-3}$.

symmetric value of 180 MeV is to be compared to the $K_0 = 210 \pm 30$ MeV deduced from the analysis of the giant monopole resonance in lead (Blaizot 1980; Blaizot, Gogny, and Grammaticos 1976), and arguments have been given (Brown 1988; Brown and Osnes 1985) that a smaller value may be more appropriate. At any rate, the value of K_0 used has little importance for the prebounce dynamics, although it is significant for postbounce dynamics.

All the models also use the value $W_s = 31.5$ MeV which, with our choice of the dependence on neutron excess of the surface energy, gives the best fit to the nuclear binding energy tables for the nucleon separation energies. The principal effect of varying W_s is to alter the free proton abundance during the infall of the star, which has a large effect on the amount of electron capture. The effect has been thoroughly discussed by Baron *et al.* (1987) and Bruenn (1989a). The choice $W_s = 31.5$ MeV is a conservative one in the sense that it is harmful to the explosion, because it leads to a large abundance of free protons which gives more electron captures and deleptonization during the collapse and hence impels the shock wave to form further in with respect to mass in the core, thereby weakening it.

Our "benchmark" model, that used in calculations 102, 103, and 106, has an iron core mass $M_{\text{iron core}} = 1.125 M_\odot$ as compared to $M_{\text{ch}}^{(a)} = 1.089 M_\odot$. The value $g_{\text{eff}} = 0.975$ was sufficient to initiate a good collapse. (This value works well for all the models except model 105, as we discuss presently.) The central electron fraction $Y_{ec} = 0.41$ is a little lower than that given by the calculations of the Woosley and Weaver presupernova calculations but matches those of Nomoto and Hashimoto (see Fig. 2). The slope of Y_e with mass is similar in all calculations. The central entropy $S_c = 0.6$ is also somewhat lower than the calculations to date have given, as can also be seen in the figure. However, it has been shown that the core masses delivered to date have been too large to explode via the direct mechanism using the best input parameters for the physics, and so we want to consider somewhat lower masses in this work and hence lower Y_e and S values. Such values could come from further electron capture reactions during the infall, which also lead to more cooling of the presupernova model.

The model used in calculation 107 is similar to that used in 102, 103, and 106. The only difference is that the M_3 and M_4 have been displaced inward by $0.05 M_\odot$ to generate a smaller iron core. Requiring $T = T_{\text{sil}}$ at M_4 leads to a lowering of the central density of the initial model. The average Y_e of this core is the same as its ancestors, but the average entropy is somewhat decreased.

Model 104 differs from the benchmark in that the entropy in the central region is somewhat decreased, while its Y_e is slightly increased, the net effect being an increase in its Chandrasekhar mass while the iron core mass has been fixed.

Model 105 represents another small core, produced by lowering Y_e while moving the entropy structure inward. This leads to only a slight reduction in M_{ch} but a larger reduction in $M_{\text{iron core}}$. It requires $g_{\text{eff}} = 0.950$ to get a good collapse (for $g_{\text{eff}} = 0.975$, the model settled into a hydrostatic state without imploding to nuclear matter density.)

Model 108 attempts to mock up the structure that would result from a star that burns silicon only in the convective core. Thus, the point M_1 has been moved out to simulate a larger convective core, and the mass points M_2 and M_3 have been made coincident. The iron core is somewhat lower than the "benchmark" model, but the Chandrasekhar mass $M_{\text{ch}}^{(a)}$ is nearly identical. The central Y_e has been kept fixed in this model.

Models 109 and 110 are a more extreme version of model 104; the entropy in the central regions has been lowered further, while the central Y_e is midway between that of the "benchmark" and 104. In order to create a model with a hot enough outer mantle to be burning silicon, the central density of this model is significantly increased over the other models in this paper. Comparing with the results of stellar evolution calculations, the central density is not excessively high.

The third equation of state parameter γ affects only the high-density equation of state and hence the dynamics from bounce onward. The BCK equation of state uses for the high-density baryonic component the following form for the pressure:

$$P_n = \frac{K_0 \rho_s}{9\gamma} \left[\left(\frac{\rho}{\rho_s} \right)^\gamma - 1 \right]. \quad (17)$$

Increasing either K_0 or γ stiffens the equation of state, and it has been shown that a softer equation of state works to strengthen the explosion (BCK; Baron 1985; Baron *et al.* 1987; Bruenn 1989b; Myra and Bludman 1989). We have used $\gamma = 3$, a stiff value, in all calculations except calculations 106 and 110, which show the effect of a softer equation of state.

Calculation 102 differs from 103 only in that it neglects neutrino-electron scattering.

VI. HYDRODYNAMICS OF MODELS

Regardless of how our initial models have been motivated and calculated in the previous sections, they can be taken as given, and then hydrodynamics can be pursued. Indeed, this is exactly what is done when the numerical simulations take as input the results of detailed presupernova investigations.

Computational details of our hydrodynamic program have been given elsewhere (Cooperstein and Baron 1989). The code uses completely general relativistic Lagrangian hydrodynamics in an energy-conserving scheme. Fine resolution is used in the regions through which the shock propagates, and shocks are modeled with the use of artificial viscosity. One or more rezonings may be made at late times in order to combine zones which have settled down and/or to further resolve zones where density gradients have become large.

Neutrino transport is also calculated with full general relativity (Baron *et al.* 1989) using a two-fluid nonequilibrium transport method (Cooperstein, van den Horn, and Baron 1986, 1987) which has been reviewed in detail by Cooperstein (1988). All types of neutrinos and antineutrinos are considered. Source terms are used for nuclear β -reactions (on free nucleons), neutrino-electron scattering, and pair and plasmon processes. Other contributions to the opacity include neutral current scattering off of nucleons and nuclei, coherent and incoherent.

In Table 3, we list the results from the calculations we have performed. We see that only models 109 and 110 have more than 10^{50} ergs of outgoing kinetic energy, although calculations 102 and 104 also have shock waves that are still propagating when the calculations were stopped, once the shocks had reached about 2000 km.

a) Model 103

Model 103 is our benchmark model, and it is constructed to be essentially a smaller version of the $1.28 M_\odot$ iron core, corresponding to a $15 M_\odot$ main-sequence star constructed by Woosley and Weaver (1988). This model has a relatively stiff high-density equation of state, and neutrino electron scattering (NES) is included in the neutrino transport. We see from Table

TABLE 3
RESULTS OF HYDRODYNAMICAL SIMULATIONS

PARAMETER	MODEL								
	102	103	104	105	106	107	108	109	110
R_{dead}	...	566.1	...	318.3	1117.3	263.2	365.5	...	...
M_{dead}	...	1.167	...	1.125	1.208	1.125	1.125	...	...
E_{ν_e}	7.08	3.39	5.92	1.16	6.18	0.87	2.27	4.69	4.61
$E_{\bar{\nu}_e}$	5.21	1.36	4.05	0.03	4.24	0.01	0.31	2.95	2.90
$E_{\nu_{\text{pair}}}$	7.13	1.98	6.20	0.20	6.17	0.13	0.73	3.96	4.64
E_{kin}	0.01	...	0.01	...	...	...	...	0.14	0.26
u_{max}	0.49	...	0.66	...	...	...	...	1.59	2.05
R_s	2104	...	1754	...	...	...	...	2682	3159
M_s	1.267	...	1.208	...	...	...	...	1.192	1.200

NOTES.—The results at the end of the calculation are described for each model. The radius (in km) and mass point (in $M_{\odot}$) at which the shock wave stalls are listed as R_{dead} and M_{dead} , respectively. The total amount of energy lost in electron type neutrinos, electron type antineutrinos, and μ and τ pair neutrinos is listed as E_{ν_e} , $E_{\bar{\nu}_e}$, $E_{\nu_{\text{pair}}}$. The energies are given in units of 10^{51} ergs. If the shock wave is still propagating at about 2000 km, the maximum outward velocity at that time (in units of 10^9 cm s $^{-1}$) is listed as u_{max} , the total out going kinetic energy is E_{kin} (in units of 10^{51} ergs), and the radius (in km) and mass point (in $M_{\odot}$) of the shock wave are listed as R_s and M_s , respectively.

3 that the shock wave just makes it to the beginning of the silicon shell at about 550 km before it stalls and becomes an accretion shock.

b) Model 102

In model 102, we study the effects of neutrino electron scattering on the infall and subsequent shock propagation. Our results confirm the work of Bruenn (1985, 1989a) and Myra and Bludman (1989). Model 102 is identical to 103 except that in model 102, NES has not been included in the neutrino transport. We see from Table 3 that model 102 produces a significantly more powerful shock wave than 103 with the shock wave still propagating at 2100 km, albeit weakening significantly. The reason for the more powerful shock is immediately evident from Figure 5, which displays the Y_l profiles of both models at bounce. We see that the effects of NES reduce the final Y_l by about 0.02, which is quite significant. Bruenn (1989a) finds that NES reduces the final Y_l by about 0.04. However,

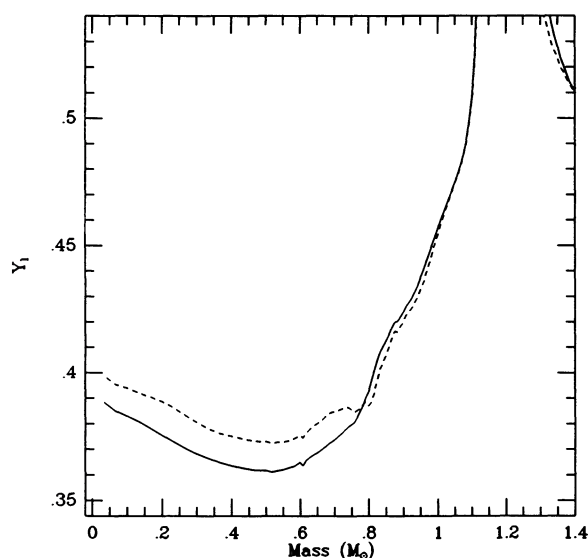


FIG. 5.—The Y_l profiles at bounce of model 103 (solid line) and model 102 (dashed line). The lack of NES in model 102 raises the final trapped lepton fraction by roughly a constant value of 0.02. This has large effects on shock formation.

with NES included, both our calculations and those of Bruenn find a total final Y_l of about 0.365, i.e., Bruenn finds significantly less capture on infall with no NES than we do. The difference is due to differing treatments of neutrino transport, but it is important to note that we agree when NES is included.

c) Model 106

Model 106 is another variation on our “benchmark” model, 103. In this calculation, we study the effects on the shock propagation of softening the high-density equation of state. Model 106 had the value of the parameter γ in equation (17) reduced from $\gamma = 3$ to $\gamma = 2.5$, producing a soft but not excessively soft equation of state. It is well known (BCK) that softening the high-density equation of state strengthens the shock wave. We see that although the shock wave is indeed strengthened, propagating out to over 1100 km versus 550 in the stiffer case (103), the shock is still not strong enough to produce an explosion. In fact, comparing 102, 103, and 106, we see that the harmful effects of NES are not completely overcome by the somewhat softer equation of state.

d) Model 104

Model 104 is significantly colder in the central regions than our “benchmark” model. This extra cooling could occur, for example, if beta decays on heavy nuclei play a more important role than has been assumed in current stellar evolution calculations (Aufderheide *et al.* 1989). The effect on the collapse of having such a low initial entropy is to reduce the fraction of free protons. This reduction in the free proton abundance substantially quenches the amount of electron capture on free protons and thus leads to a higher final trapped lepton fraction at bounce as displayed in Figure 6, where the lepton fraction is compared to that of model 103. However, whereas the previous models that we have considered are both hot enough and have a low enough initial Y_e that the effects of electron capture on heavy nuclei would probably not give a large contribution to the total reduction in the final trapped lepton fraction, the initial Y_e is high in this model. The available nuclei have states for which allowed transitions can proceed, and because there are very few free protons, electron capture on nuclei could become a more important source of lepton number reduction. In these calculations, we have not included the effects of such captures on heavy nuclei, and so we must temper our conclu-

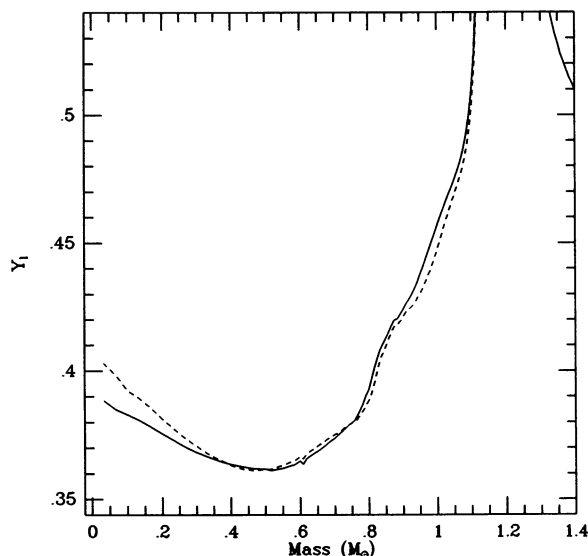


FIG. 6.—The Y_l profiles at bounce of model 103 (solid line) and model 104 (dashed line). The colder initial model 104 results in reduced electron capture in the center; this in turn leads to a stronger shock wave.

sions until this process is properly included in such cold models. In any event, it is clear that this model produces a weak explosion; while not strong enough to actually be a supernova, it shows that such a cold core is helpful.

e) Models 105 and 107

In models 105 and 107, we examine the effect of making the iron core smaller simply by altering the positions of the silicon burning shells, that is, by moving them in. The results are interesting and perhaps somewhat paradoxical. As displayed in Figures 7 and 8, the trapped lepton fraction at bounce is significantly lower in these models than for model 103. Both the shock waves make it out of the putative iron core and die, in fact, at the same mass point $1.125 M_\odot$. Specifically, the impact

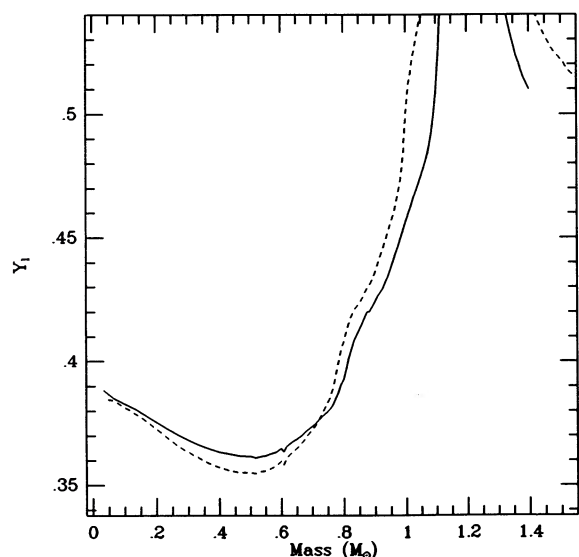


FIG. 7.—The Y_l profiles at bounce of model 103 (solid line) and model 105 (dashed line). Even though the “iron core” of model 105 is reduced from that of 103, there is significantly lower Y_l in 105 than 103.

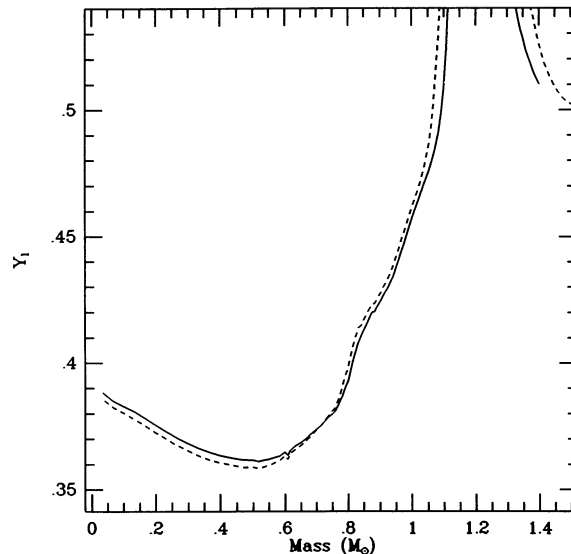


FIG. 8.—The Y_l profiles at bounce of model 103 (solid line) and model 107 (dashed line). This situation is similar to that of model 105, with a smaller iron core leading to a lower final trapped lepton fraction.

that changing the location of the burning shells has on the final trapped lepton fraction is not obvious. This reduction in the lepton fraction does not occur until the center reaches about $10^{13} \text{ g cm}^{-3}$, so it must alter the flow of neutrinos, but exactly how this occurs is not clear, and we have not yet arrived at a satisfactory explanation of this complicated behavior. Comparing the initial density profiles (Fig. 9), the models appear more favorable than 103, yet they fare worse. One might note that the densities are flatter in models 105 and 107 at the mass point $1.125 M_\odot$, where the shock stalls, which is harmful to the shock; however, the models have much steeper density profiles earlier in mass which should have given the shock a bit of a push, so it is not clear that the density structure is responsible.

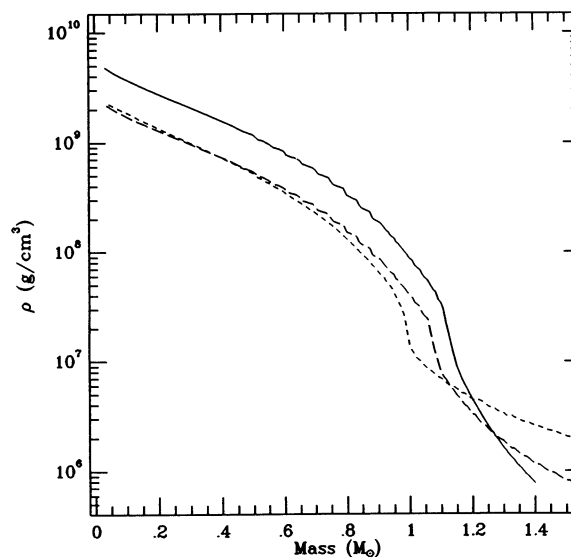


FIG. 9.—The initial density profiles of model 103 (solid line) compared to that of models 105 (short dashed line) and model 107 (long dashed line). Models 105 and 107 look as though the density profiles should be more favorable to the propagation of the shock, with the density dropping off sooner in mass, but the shocks do not get as far in mass or radius before stalling.

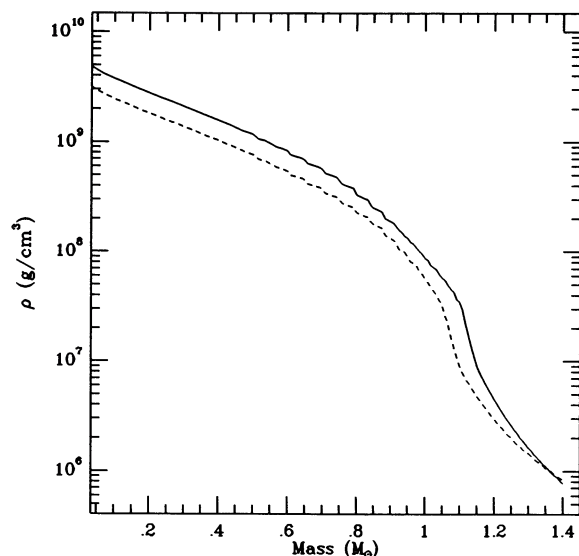


FIG. 10.—The initial density profiles of model 103 (solid line) and model 108 (dashed line).

f) Model 108

Model 108 mocks up an iron core where there was no silicon shell burning prior to iron core collapse. This model is similar to that of the $13 M_{\odot}$ star ($1.18 M_{\odot}$ iron core) of Nomoto and Hashimoto (1988); the initial density profile is shown in Figure 10. The results of this model are very similar to that of models 105 and 107. As seen in Figure 11, the lepton fraction at bounce is again reduced over that of model 103, and the shock wave dies at the same mass point. Again, it is unclear why altering the outer structure alters the trapped lepton fraction inside the homologous core. It is important to point out that when the $1.18 M_{\odot}$ star of Nomoto and Hashimoto (1988) is compared to the $1.28 M_{\odot}$ iron core of Woosley and Weaver (1988), similar results obtain (Baron 1988; Cooperstein and Baron 1989): the smaller iron core is harder to explode. In both

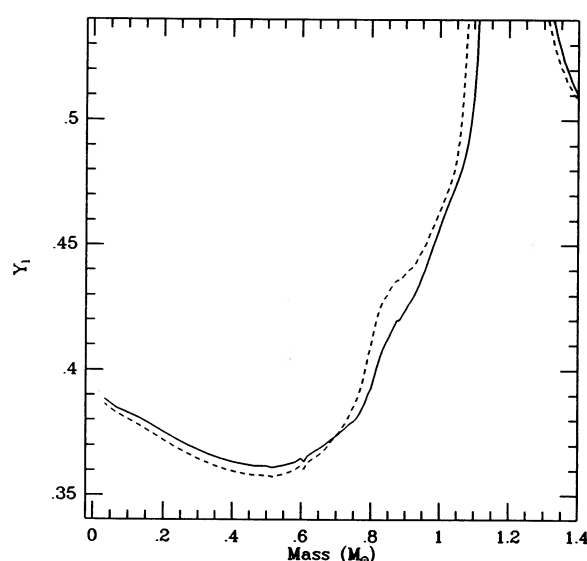


FIG. 11.—The Y_e profiles at bounce of model 103 (solid line) and model 108 (dashed line). This situation is similar to that of model 105, with a smaller iron core leading to a lower final trapped lepton fraction. The average entropy initially is lower in model 108 than in 103, leading one to expect the reverse situation.

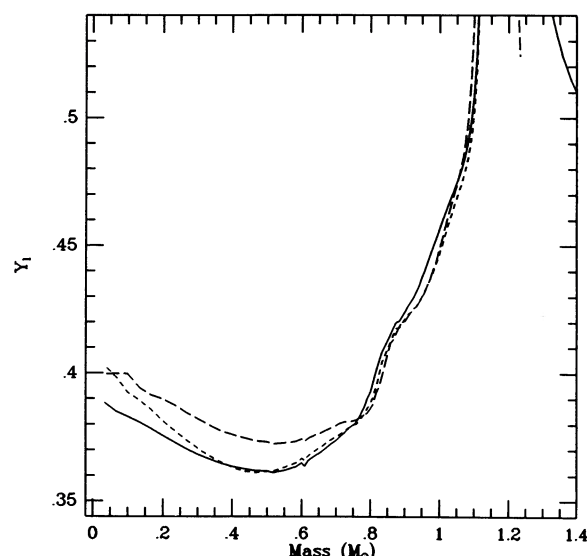


FIG. 12.—The Y_e profiles at bounce of model 103 (solid line), model 104 (short dashed line), and model 109 (long dashed line); the colder models have less electron capture on infall, leading to shock formation at a larger mass point. The central entropy is the same in models 104 and 109, but the average entropy inside the mass point M_1 is lower in 109, which further increases the average Y_e . This in turn strengthens the shock considerably.

cases, the initial density profiles of the smaller cores look more favorable.

g) Models 109 and 110

These models represent more extreme versions of model 104. The mass of the iron core is kept fixed, as is the central entropy. It is the slope of the entropy in the initially convective core that is reduced. Figure 12 shows the Y_e at bounce for model 109 compared with models 103 and 104. The average Y_e is substantially higher in model 109. This is reflected in the strength of the shock wave. Figure 13 displays the velocity profiles of

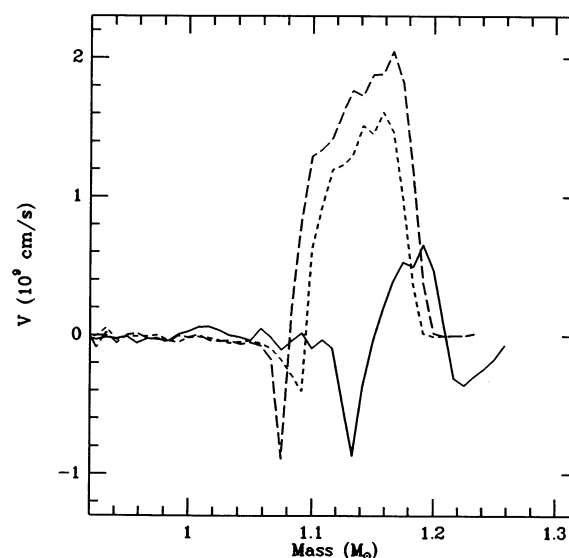


FIG. 13.—The final velocity profiles of models 104 (solid line), 109 (short dashed line), and 110 (long dashed line). All these models have shock waves that are still propagating at the edge of the numerical grid, but the colder model 109 is significantly stronger with a total energy in the shock wave being about 0.5×10^{51} ergs. Model 110 has a softer equation of state and a more powerful shock wave, with an energy of about 1.3×10^{51} ergs.

models 104, 109, and 110 at the end of the calculations. Models 109 and 110 have reached the end of the computational grid with substantial velocities. An estimate of the total energy in the shock wave for model 109 is about 0.5×10^{51} ergs and 1.3×10^{51} ergs for model 110. Model 110 is stronger simply because the high-density equation of state was softer for that model. In these models, the same caveat about electron capture on heavy nuclei that we noted in our discussion of model 104 applies. It is still gratifying to see that the prompt mechanism can work on some models with conservative input physics for the equation of state and the symmetry energy.

VII. DISCUSSION AND CONCLUSIONS

The above hydrodynamical results seem to show that the single most important factor determining the success or failure of the shock wave is the value of the final trapped lepton fraction Y_l in the homologous core of the star. The models that are cold initially and hence are subject to less electron capture on infall produce viable shock waves. These models are uncertain, however, as to the effects of capture on heavy nuclei as discussed above. A variety of agencies may affect the final Y_l , including the outer regions of the core, in a complicated fashion. Regardless of the actual cause of the variation, small changes in Y_l lead to large changes in the strength of the shock. This is primarily because the shock formation point moves inward with decreasing Y_l (Burrows and Lattimer 1983; Cooperstein, Bethe, and Brown 1984) as has been well studied.

Our results clearly show that the mass of the iron core, at least as we have defined it, does not allow one to easily predict the effect on the shock wave when compared with some other

core. In fact, our calculations seem to show that there is a minimum mass below which the effects of reducing the mass still further become detrimental. This can be seen in Table 3 by comparing our “benchmark” model 103 with models 105, 107, and 108 which all have smaller iron cores; the shock is weaker in these models. We have no unambiguous explanation for this effect, and the possibility remains that it is the result of our assumptions, although as we noted in the previous section, the same effect occurs with models that come from the results of stellar evolution calculations. It appears that it is harmful to have too large a reduction in the iron core mass below the Chandrasekhar mass. It is striking from Table 1 that the “best” models, 104, 109, and 110, have $M_{\text{iron core}}$ slightly greater than M_{ch} . It would seem that it may be possible that iron cores with masses below about $1.1 M_{\odot}$ could produce strong shock waves, but such models would have to look very different in their entropy and composition structure than those we have considered.

We wish to thank Morry Aufderheide for allowing us to use the results of our joint unpublished work on neutrino electron scattering, Paul Schinder for providing us with his results on plasmon emission rates, Ken Nomoto and Stan Woosley for providing us with the results of their calculations and useful discussions, Hans Bethe for helpful discussion, and the referee for suggestions clarifying the presentation. The calculations reported herein were performed at the National MFE computer center under auspices of the Nuclear Physics division, US Department of Energy.

APPENDIX

Evaluating perturbative corrections to the Chandrasekhar mass has been done in some detail in the literature (see Shapiro and Teukolsky 1983), but it is important to do this specifically for the equation of state and density regime encountered in supernova material beginning to collapse.

A completely relativistic gas follows the equation of state

$$P = \kappa \rho^{4/3}. \quad (\text{A1})$$

For the zero-temperature massless electron gas,

$$\kappa = \frac{1}{3\pi^2} \frac{k_e^4}{4} \left(\frac{m_n}{\rho} \right)^{4/3}, \quad (\text{A2})$$

where the electron fermi-momentum is given by

$$k_e = \left(\frac{3\pi^2 \rho Y_e}{m_n} \right)^{1/3} = 0.045 \left(\frac{Y_e \rho_{10}}{0.50} \right)^{1/3} \text{ fm}^{-1} = 8.82 \left(\frac{Y_e \rho_{10}}{0.50} \right)^{1/3} \frac{\text{MeV}}{\hbar c}. \quad (\text{A3})$$

Many textbooks show that the limiting mass for this equation of state is given by

$$M_{\text{ch}} = 4.555(\kappa/G)^{3/2}, \quad (\text{A4})$$

in cgs units when κ is defined assuming ρ is the mass density. Using the value of the nucleon mass m_n , appropriate to iron one finds the limiting mass to be

$$M_{\text{ch}}^{(a)} = 1.44 \left(\frac{Y_e}{0.50} \right)^2 M_{\odot}. \quad (\text{A5})$$

(In the literature, one sometimes sees the constant 1.44 replaced by values up to 1.46, reflecting a different choice of nucleon mass.)

If the equation of state is not purely of the form equation (A1) but is dominated by $\gamma = 4/3$ terms, then one may consider perturbations to the limiting mass due to other terms in the pressure. Modifications to gravitation due to general relativity and rotation may also be taken into account perturbatively. The equation of state effects are most conveniently evaluated by assuming

$$M_{\text{ch}} = (1 + \delta M) M_{\text{ch}}^{(a)}, \quad (\text{A6})$$

where we obtain from equation (A4)

$$\delta M = \frac{3}{2} \left(\frac{P - P_{e0}}{P_{e0}} \right), \quad (\text{A7})$$

with P_{e0} being the pressure of the massless electron gas at $T = 0$. This procedure takes into account behavior other than $\gamma = 4/3$ in the equation of state by altering the value of κ instead. Equivalently, one could find the limiting mass for any equation of state by maximizing the mass with respect to central density, as we did to obtain $M_{\text{ch}}^{(c)}$ in § III, but this requires numerical work.

The corrections to P_{e0} in supernova matter come from the finite temperature and mass of the electrons, the contributions of baryons, including Coulomb lattice pressure and ionic pressure, and radiation. The lattice pressure involves both electrons and the baryons, but we will take it as a baryonic contribution. We first consider the electronic corrections.

For a free electron gas of nonzero temperature and mass, the electron density and pressure are given by

$$\begin{aligned} \rho Y_e &= \frac{1}{\pi^2} \int_0^\infty dk k^2 f_e, \\ P_e &= \frac{1}{3\pi^2} \int_0^\infty dk k^3 \frac{dE}{dk} f_e, \end{aligned} \quad (\text{A8})$$

where

$$f_e = \left\{ 1 + \exp \left[\frac{E(k) - \mu_e}{T} \right] \right\}^{-1}; \quad E^2(k) = m_e^2 + k^2. \quad (\text{A9})$$

We can rewrite these expressions as

$$\rho Y_e = \frac{1}{\pi^2} \int_{m_e}^\infty dE E (E^2 - m_e^2)^{1/2} f_e, \quad (\text{A10})$$

$$P_e = \frac{1}{3\pi^2} \int_{m_e}^\infty dE (E^2 - m_e^2)^{3/2} f_e. \quad (\text{A11})$$

Evaluating these integrals to quadratic order in $(\pi T/\mu_e)$ and (m_e/μ_e) , we obtain

$$\begin{aligned} \frac{1}{\pi^2} \frac{k_e^3}{3} &\equiv \rho Y_e = \frac{1}{\pi^2} \frac{k_e^3}{3} \left[1 + \left(\frac{\pi T}{\mu_e} \right)^2 - \frac{3}{2} \left(\frac{m_e}{\mu_e} \right)^2 \right], \\ P_e &= \frac{1}{3\pi^2} \frac{\mu_e^4}{4} \left[1 + 2 \left(\frac{\pi T}{\mu_e} \right)^2 - \frac{3}{4} \left(\frac{m_e}{\mu_e} \right)^2 \right]. \end{aligned} \quad (\text{A12})$$

Using

$$\mu_e \simeq k_e \left[1 - \frac{1}{3} \left(\frac{\pi T}{k_e} \right)^2 + \frac{1}{2} \left(\frac{m_e}{k_e} \right)^2 \right], \quad (\text{A13})$$

we find for the pressure

$$P_e = \left[1 + \frac{2}{3} \left(\frac{\pi T}{k_e} \right)^2 - \left(\frac{m_e}{k_e} \right)^2 \right] P_{e0}; \quad P_{e0} = \frac{k_e^4}{12\pi^2}. \quad (\text{A14})$$

Since the entropy per nucleon in electrons is given by $S_e = Y_e \pi^2 T/k_e$, we can rewrite this as

$$\frac{P_e}{P_{e0}} = 1 + \frac{2}{3} \left(\frac{S_e}{\pi Y_e} \right)^2 - \left(\frac{m_e}{k_e} \right)^2, \quad (\text{A15})$$

which leads to a fractional change in M_{ch} given by

$$\delta M_{\text{electrons}} = \frac{3}{2} \left(\frac{P_e - P_{e0}}{P_{e0}} \right) = \left(\frac{S_e}{\pi Y_e} \right)^2 - \frac{3}{2} \left(\frac{m_e}{k_e} \right)^2. \quad (\text{A16})$$

The term due to finite mass is quite small at the central density of the presupernova core; for $Y_e = 0.50$ at $\rho_{10} = (1., 0.1)$ we find $3/2(m_e/k_e)^2 = (0.00504, 0.0234)$. At lower densities, such as those found at the outer edge of the iron core, the correction is more important, but following the same argument we used in the main text to use S_e near the center, it is the central values which are important for determining the Chandrasekhar mass.

We now consider the baryonic contributions beginning with the Coulomb lattice pressure. In general, this is complicated to compute exactly if the material consists of a nuclear statistical equilibrium ensemble of different nuclear species, but the Wigner-Seitz approximation gives sufficiently accurate results. In this approximation, one considers an electrically neutral sphere containing one large nucleus and computes the electrostatic Coulomb energies due to attraction of the nuclei to the electron background and

the repulsion of the electrons for each other and the protons within the nucleus for each other (Baym, Bethe, and Pethick 1971; Bethe *et al.* 1979). One must also compute the surface energy of the nucleus, because the competition between surface and Coulomb energies determines the optimal nuclear size. Using the compressible liquid drop model, one can determine the pressure which arises from the Coulomb and surface energies. The result is (Cooperstein 1985; Cooperstein and Baron 1989)

$$\frac{P_{\text{size}}}{\rho} = 0.0416(1 - 0.0062T^2)\rho_{10}^{1/3} \text{ MeV} . \quad (\text{A17})$$

The small T^2 factor arises from the reduction of the nuclear level densities that occurs as the average A values of the nuclei increase with rising baryon density. We find

$$\frac{P_{\text{size}}}{P_{e0}} = 0.0377(1 - 0.0062T^2) , \quad (\text{A18})$$

which gives for the correction to the Chandrasekhar mass

$$\delta M_{\text{lattice}} = 0.057 , \quad (\text{A19})$$

where we drop the temperature correction. Note that there is no density dependence in this correction; i.e., the lattice pressure follows $\gamma = 4/3$ behavior.

The remaining correction due to nuclei is that due to the ionic ideal gas pressure. The pressure due to ions is simply

$$\frac{P_{\text{ion}}}{\rho} = \left\langle \frac{1}{A} \right\rangle T , \quad (\text{A20})$$

where $\langle 1/A \rangle$ is the average reciprocal atomic number of the nuclei. We find

$$\delta M_{\text{ion}} = \frac{3}{2} \left(\frac{P_{\text{ion}}}{P_{e0}} \right) = \frac{3}{2} \left(\frac{\langle 1/A \rangle T}{Y_e k_e/4} \right) , \quad (\text{A21})$$

and putting in numbers

$$\delta M_{\text{ion}} = 0.680 \left\langle \frac{1}{A} \right\rangle \rho_{10}^{-1/3} T = 1.21 \left\langle \frac{1}{A} \right\rangle S_e , \quad (\text{A22})$$

where the second form is convenient.

The final contribution to the pressure is due to radiation. It is trivial to show that

$$\frac{P_{\text{rad}}}{P_{e0}} \sim \left(\frac{T}{k_e} \right)^4 \sim 0 , \quad (\text{A23})$$

and thus it does not need to be included.

We now consider the two final corrections due to non-equation of state sources, due to general relativity and rotation. General relativity raises the critical adiabatic index above $4/3$. On quite general grounds, we can state

$$\delta M_{\text{GR}} \sim \frac{GM}{Rc^2} \sim \frac{P_c}{\rho_c c^2} . \quad (\text{A24})$$

Putting in numbers, we find

$$\frac{GM}{Rc^2} \sim 0.0015 \left(\frac{M}{M_\odot} \right) \left(\frac{R}{1000 \text{ km}} \right)^{-1} , \quad (\text{A25})$$

$$\frac{P_c}{\rho_c c^2} \sim \frac{Y_e k_e/4}{m_n} \sim 0.00125 \rho_{10}^{-1/3} , \quad (\text{A26})$$

and so it is safe to ignore the general relativistic correction. The effects of rotation can be estimated by assuming

$$\delta M_{\text{rotation}} \sim \frac{\omega^2 R^2}{GM/R} ; \quad (\text{A27})$$

even for a star spinning rapidly enough to become a submillisecond pulsar, this is at most a 1% correction, and so we will neglect it.

In summary, we find

$$\delta M = -0.057 + \left(\frac{S_e}{\pi Y_e} \right)^2 + 1.21 \left\langle \frac{1}{A} \right\rangle S_e . \quad (\text{A28})$$

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E. BARON: Department of Physics, State University of New York, Stony Brook, NY 11794-3800

J. COOPERSTEIN: Department of Physics, University of Michigan, Ann Arbor, MI 48109